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The antibody revolution is real. The business, less so.

7.1k words, 28-35 min reading time

CARLOS OUTEIRAL
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I have been meaning to write about the recent antibody revolution. The problem is, you see, that I wanted to make sure it was real. **AI for drug discovery has tended to experience revolutions the way other industries have quarterly results: four times a year, always the one that will finally change everything.** After a decade in the field, you build a reflex: you read the paper, then look for the catch.

Take the 2023 antibody “revolution”. Reports of the first [“de novo designed” AI antibody](#) by [AbSci](#) made it to [international newspapers](#) and created a frenzy on social media. The catch? They made a variant of [trastuzumab](#), an antibody targeting [HER2](#) in breast cancer, **approved in 1998**, with just one loop replacement. Also, it was not really *de novo*: they screened **half a million variants**, to find only three winners... the bar being, again, improving on a twenty-five year old molecule! The experimental design also had [clear caveats](#), including lack of appropriate controls. But despite all of this, the [press release](#) read like the biology equivalent of the moon landing.

Then, last year, not one but **four companies** again announced the ability to design *de novo* antibodies using AI/ML: [Chai Discovery](#), [Nabla Bio](#), [Latent Labs](#) and Xaira Therapeutics (though licensing work from [David Baker’s lab](#)). If you remember the trastuzumab fiasco, you surely can’t blame me for facing the news with skepticism. One of the founders of Chai, Joshua Meier (a solid researcher originally from the Meta team that developed the popular [protein language model ESM](#)) was also the corresponding author on the infamous AbSci technical report. There is something new, though. **This time, the different labs have all reported novel antibodies against multiple, novel targets**, with little to no optimization. The hit rates are now in the double digits, and the overall pharmacological properties of these antibodies start to look like proper drugs. And an increasing number of offshoot competitors, and even some academic groups, are reporting similar results.

This time, I think, the science is real. But the excitement it deserves has been buried under hype from an AI-heavy commentariat that has mostly not bothered to ask what it means for drug discovery. Some of the things they are missing are:

- why the *de novo* antibody discovery technology is a genuine scientific achievement
- why it might not necessarily result in the industry making better drugs
- why it probably is not a good business, unless these companies think very carefully about how to create value

I would like to address this today.

Why we care about antibodies

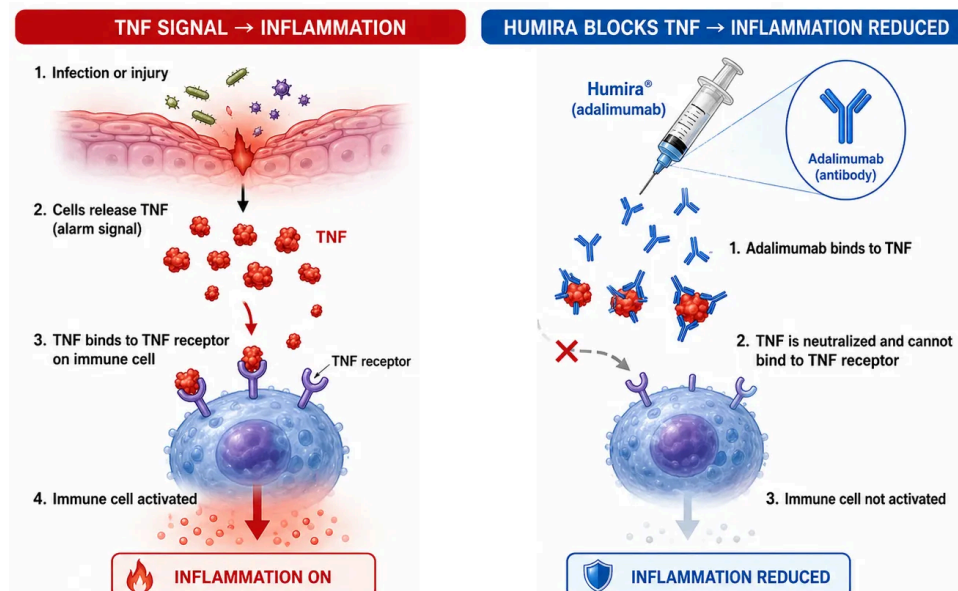
Most of the excitement rests on an assumption that seems obvious if you haven't been in the field before: that designing an antibody and designing a drug are the same problem. That thesis falls right at the outset. While it is true that antibodies *can be* drugs, or perhaps more accurately, that *some drugs* are antibodies, most antibodies are *not, and could not, be drugs*. The argument is not trivial, and my best chance to get it through is by starting with what an antibody actually is.

[Antibodies](#) are natural proteins expressed by your immune system to fight off invaders. They are frankly incredible, the byproduct of millennia of evolution to produce a lean, mean, defensive machine that can recognise and inactivate a variety of threats. For this essay, what matters most is that **an antibody can recognise a specific target protein with incredibly strong affinity while leaving almost anything else untouched**. If you have ever taken a [pregnancy test](#) (or a CoVID lateral flow test), you have been using a strip impregnated with an antibody that immediately latches on to the target protein while ignoring everything else. This [magic bullet](#) property is also what has made them incredibly promising as drugs.

The implication of an antibody drug is that a specific (or, as we say, [monoclonal](#)) antibody can be designed and injected to attack some proteins in the human body. Any proteins? Not really. Because they are large and bulky, antibodies cannot cross the membrane that separates a cell from its surroundings, so they are limited to act on proteins displayed on the cell's exterior (there are [some variants](#) that can get inside the cell, but they are still prototypes, and none have been approved by the FDA yet). Limiting to extracellular proteins is fine because they are roughly one fifth of all human proteins, and they are disproportionately represented in disease.

How do these antibody drugs work? Take Humira ([adalimumab](#)), one of the largest-grossing drugs in history. Humira acts as a *protein-protein interaction inhibitor* for a protein called [tumor necrosis factor](#) (TNF). TNF is an alarm signal for the cell: whenever there is an infection or injury, cells release it to switch on inflammation, recruiting the immune system and opening the body's resources to combat damage. This is usually a good thing. But if you have an autoimmune disease like rheumatoid arthritis, or Crohn's disease, or psoriasis, or many others, your body is stuck generating TNF and turning on itself, and switching it off has therapeutic value.

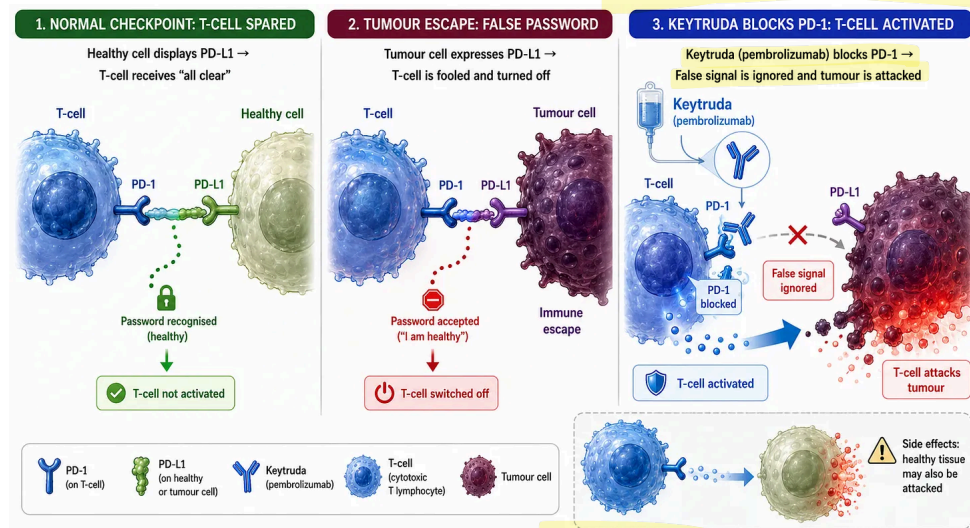
And that's exactly what happens when you get a Humira injection. The antibody latches onto the TNF protein and coats it, preventing it from reaching the immune system cells that it would otherwise activate and switching off the autoimmune disorder. Because of the generality of the mechanism, the enormous patient population and (let's not beat around the bush) the fact that it needs to be taken regularly because autoimmune diseases are chronic, Humira became one of the best selling drugs in history, peaking at \$20B/year before it went off-patent in 2023.



Infographic describing the mechanism of Humira (adalimumab). The antibody binds free tumor necrosis factor (TNF) in the bloodstream, blocking it from binding the TNF receptor on immune cells and inducing inflammation. Infographic lightly edited from a GPT Image-2 generation, June 2026.

Since we are talking about Humira, it may be worth talking about the king of antibodies, the one that will figure in the pitch deck of any AI for antibody discovery company. Keytruda ([pembrolizumab](#), ~\$32B in revenue in 2025) targets a protein called [PD-1](#), which is expressed by T-cells, specialised cells of the immune system that hunt down other abnormal cells, like tumour cells. PD-1 works like a password recogniser: it inspects healthy cells for another protein called [PD-L1](#) that flags them as healthy. Unfortunately, tumour cells learn this trick quite quickly and express PD-L1 on their surface to pass the check. Keytruda blocks PD-1 so the false password is never read, and the T-cell attacks the tumour anyway.

If you are paying attention, you may now be wondering why these T-cells do not attack healthy cells. Well, they do. It is one of the major side effects. But, as it often happens in drug discovery, smart engineering can adjust a molecule to produce a [very large effect on diseased cells while only barely inconveniencing healthy ones](#). In effect, the Phase III clinical trials for Keytruda (first for [melanoma](#), then for many others) showed a doubling in the overall survival, with respect to the standard of care, and with relatively minimal side effects.



Infographic describing the mechanism of Keytruda (pembrolizumab). The antibody binds the PD-1 receptor on the surface of T cells, forcing them to destroy tumour cells. Infographic generated using GPT Image-2, June 2026.

More broadly, there is a reason to care about antibodies that goes beyond any single drug: they help us push back against [Eroom's law](#), the observation that drug discovery is becoming slower and more expensive over time. Antibodies have significantly higher success rates than small molecules. One of the most extensive studies found that they are around 50% more likely to be approved (12% vs 8% Phase I-to-approval), which is driven mostly by success at Phase II/III and regulatory review. In other words, an antibody program is far likelier to reach patients than a small-molecule one, and development times have also [shortened significantly](#).

Antibodies also come in increasingly sophisticated variants. [Bispecific antibodies](#), for example, are designed to have different arms with different specificity for different proteins. Perhaps the best example is [emicizumab](#), which brings together two factors in the coagulation cascade and which has been transformative in the treatment of hemophilia. And the last few years there have seen a frenzy around [antibody-drug conjugates](#), which are antibodies with a cytotoxic attachment, where the antibody identifies a cancerous cell and delivers their payload. But the near totality of work in the AI/ML field has focused on two types of antibodies, [nanobodies](#) and [single-chain variable fragments](#) (scFvs), that follow the principles we discussed above.

Let's leave it at this: antibodies are protein drugs that target proteins on the cell's exterior, with high specificity, the ability to reach undruggable targets, and better clinical success rates than small molecules. They are, frankly, remarkable. But there are some caveats.

What it entails to make an antibody

I am saying a lot of great things about antibodies here, which is not really my style. I am leaving aside several problems here for the sake of brevity, although they are extraordinarily important: the cost of manufacturing (roughly ~100x more expensive than small molecules), the supply chain issues (need to be refrigerated) and the pharmacology problems (can't be orally dosed, need to be injected/infused or in the best case administered by sub-cutaneous injection, which is painful and inconvenient).

None of these are trivial for the ultimate mission of medicine, which is to give universal, affordable care to everyone.

But there is one really great thing about antibodies: *immune systems know how to make them*. Let me explain. In the realm of [small molecule drugs](#), where most of pharma still spends most of their time, there are rarely guarantees of success. Many targets are [undruggable](#), which means they are proteins without a [druggable pocket](#) where a small molecule can bind. Some of these undruggable targets ([Myc](#), [β-catenin](#), [α-synuclein](#), amongst many others) have been known for decades and we are no closer to find an effective drug for them than we were fifty years ago. In contrast, it is almost guaranteed that given some extracellular target, you can make some antibody that binds to it.

The most common technology for antibody design can be summarised in: find a way to expose an organism to some foreign protein, and wait until their immune system engineers an antibody against it. The classical approach is the [hybridoma technology](#), which relies on a simple idea: take an animal (generally a mouse, or a rabbit), inject it with the protein you want to raise antibodies against, then look at the antibodies that the animal generated and isolate one. In recent years there have been a few improvements, such as using [transgenic humanised mice](#) approaches where the immunoglobulin genes from the mice have been replaced with human ones. This has the advantage that the developed antibodies are more human-like, and therefore less likely to have side effects such as [anti-drug antibodies](#). While it has some limitations (for example, you generally can't make an antibody against an endogenous protein), it is an overwhelmingly successful technology for antibody discovery.

Of course, rather than injecting a foreign entity, you can also just look at humans that have naturally evolved ones. The whole field of [natural immune repertoire analysis](#) relies on this, having led to multiple [therapies for infectious disease](#). Alternatively, you can use something called [phage display](#), or its close cousins [yeast](#) and [ribosome display](#), to test hundreds of millions of randomly generated antibodies to see if one of them binds the target. In these display technologies, antibody fragments are expressed fused to a coat protein on the surface of an organism, with the encoding gene packaged inside the particle (cell or phage). Modern setups can screen around one trillion antibodies in around two weeks.

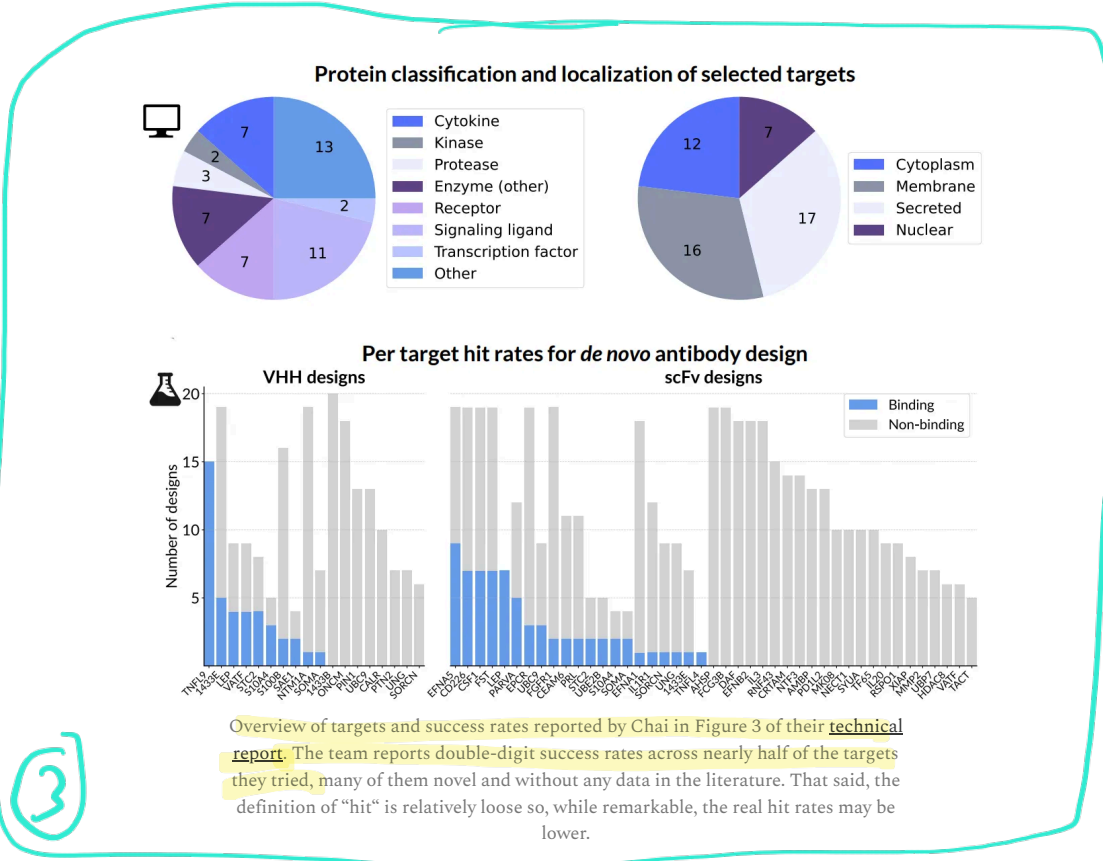
There are lots of caveats, the main one being: the antibodies generated by these methods are *binders*, and a lot of work may be required to turn them into drugs. We will come back to this. But for now, let's think about how AI/ML methods can change the way we come up with novel antibodies.

The new era of antibody design

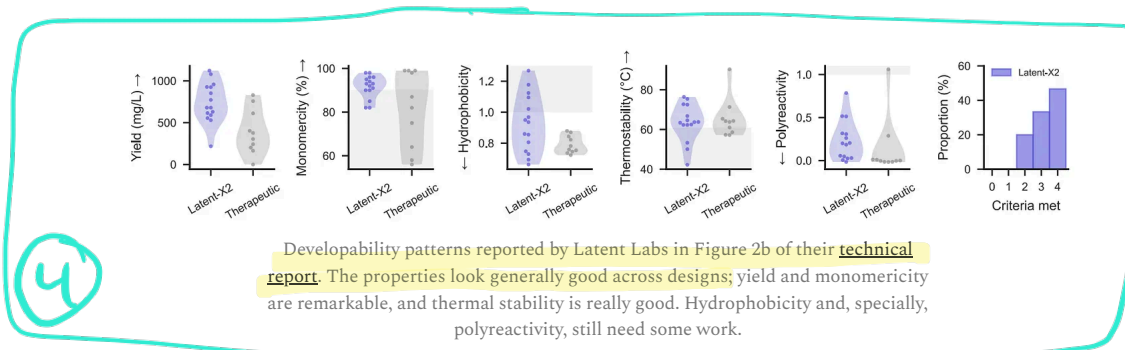
Let's go back to the breakthroughs of 2025. Four frontier BioAI companies (Nabla.Bio, Latent Labs, Chai Discovery, and Xaira Therapeutics, the latter under the wing of [David Baker](#)'s lab) reported models capable of designing antibodies *de novo*, with high-single to low-double digit success rates.

The sequence started with a [paper from David Baker's lab](#) (preprinted in March 2025, published in Nature in October 2025), presenting the technology that was licensed to Xaira Therapeutics and likely was behind their [billion dollar seed round](#). Their paper

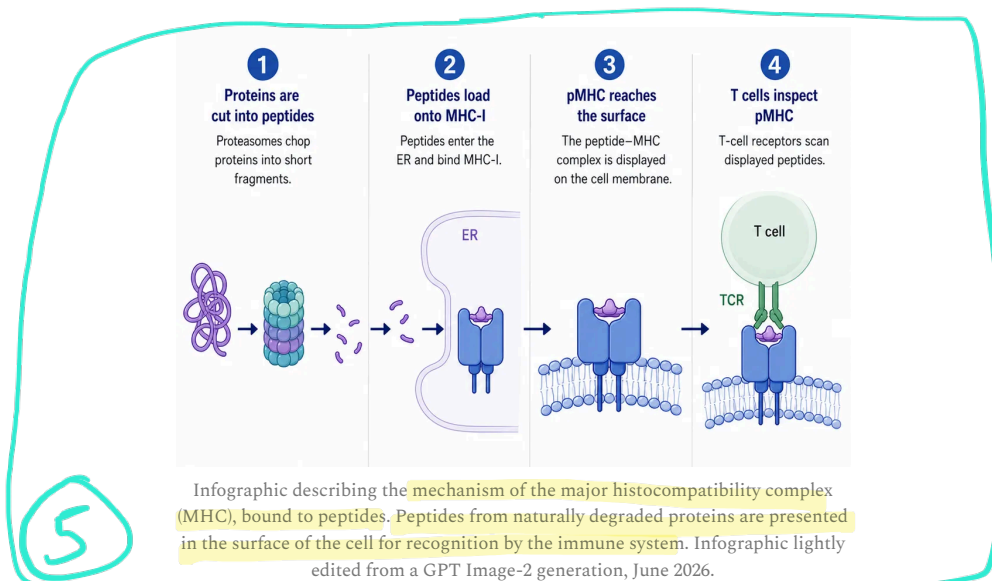
was the first demonstration that you could find a binder from scratch at all: zero-shot and against real, but well-trodden targets (hemagglutinin, toxin B from *C. difficile*, and others), although with relatively weak potencies (single-digit micromolar, or triple-digit nanomolar) and a hit rate around 1-2%. Just three months later, Chai Discovery published a [technical report](#) showing much higher-quality zero-shot antibodies, this time against unprecedented targets, with much higher potencies (generally single-digit to double-digit nanomolar). The hit rate was also much higher, reported as double digit, although of course that depends on the definition of “hit” (which here, without in any way diminishing their accomplishment, was generous).



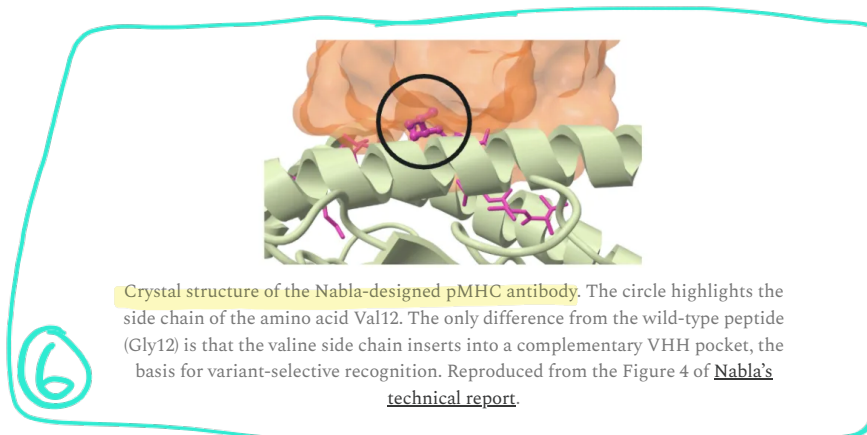
The Chai technical report was impressive, but the story doesn't stop there. Towards the end of the year, Nabra.Bio released [another technical report](#) where they showed an improved model, with similar hit rates to Chai-2, but with more rigorous benchmarking, and succeeding at some of the targets where their competitor had failed. They also reported not just antibody *binders*, but they looked at many of the properties that make an antibody a good drug (we'll talk about this in a second). For example, they report wet lab assays for thermal stability, monomericity, expression titer, and many others, where the antibodies look reasonably attractive. And if this sounds amazing, then it was topped by Latent Labs, right before Christmas, with another [technical report](#), showing similar results to other labs but with a pretty extensive amount of developability data, including even a demonstration that their antibodies had low immunogenicity when tested on T cell samples from healthy volunteers (although of course this is poorly predictive of what might happen in a real patient).



While I was editing this essay, Nabla published [yet another technical report](#) where they used their JAM-2 model to design antibodies against [peptide-MHC \(pMHC\) complexes](#). Some background will be useful here: vertebrate cells are continuously breaking apart some of their proteins, and taking small fragments (peptides) that are presented in the surface of the cell via the major histocompatibility complex (MHC). In turn, these complexes are recognised by cells of the immune system, which identify potentially foreign cells and destroy them. This system, however, is not perfect. For example, mutations in [KRAS](#), a protein that is mutated in 20% of human cancers and has been one of the most coveted drug targets for decades, do not elicit an immune response because a single change is often insufficient to make the peptide-MHC complex reliably distinguishable as “foreign” by T cells.



And here is the power of Nabla’s model: they managed to design one antibody that can selectively recognise pMHC complexes of two mutants of KRAS (G12V and G12C, the second and third most common) while leaving wild-type KRAS essentially untouched. This is an incredible achievement for two reasons. One, they are the first report, as far as I know, of a computationally designed pMHC-antibody. This is really hard, because binding a peptide is much harder than binding a globular protein, because of its flexibility and because one needs to engage specific atoms. Making it selective is even harder. But, second and perhaps more importantly, this opens the possibility of designing antibodies against intracellular proteins (with lots of caveats like HLA-restriction, the low density of the pMHC on the surface, and the fact that the broader TCR-mimic field has a graveyard all its own), which opens many doors in diseases with highly unmet medical need.



The experimental data released by all of these labs is, frankly, one of the most thorough technological demonstrations I have seen in some time. In AI for science, the proof is in the pudding, and **generating novel chemical matter for a novel target that other people have been trying to solve for a long while is as high a bar as you can possibly reach for**. If you do it for a dozen, previously unsuccessful targets, well, that is even more impressive! **While ignoring the caveats** (sure, their hit rates will be lower once you discard the weak signals, or the sensograms that look weird), **I think that this is an extremely solid demonstration that the technology is working**. It is also quite impressive to see the field advance in real time throughout two years, as the technology improved and as the bar to make a splashy announcement kept moving higher and higher.

We do not know all the details of the private models because they are, well, private. We can however infer some information from the paper by David Baker's lab, which I had the pleasure to peer-review for Nature. At first glance, pure architectural innovations in David Baker's paper are relatively modest. They use their celebrated diffusion model [RFdiffusion](#), together with some fine-tuning on published antibody structures, to design a large number of structural decoys. These decoys have a structure, but not a specific set of amino acids (i.e. they design only the [alpha-carbons](#) and omit side chains), so the [ProteinMPNN](#) neural network is used to generate a protein sequence that may fold into that structure. Finally, they used a fine-tuned version of their [RoseTTAFold 2](#) model (which you can understand as an alternative to AlphaFold 2) to predict the structure of the complex, as well as the probability of binding using a specially designed and trained head, to validate that the generation is correct. There are lots of small technical details to how this worked in practice, but the general pipeline follows the [central dogma of protein design](#) or RFdiffusion -> ProteinMPNN -> AlphaFold 2.

Buried in the supplementary methods, however, are some proprietary datasets, like the Baker Lab's entire database of yeast display experiments (1.6M measurements against 41 proteins). Data collection has become an explicit engine for some of these companies, particularly Nabla.Bio. Others appear far more computational: as far as I know, neither Chai Discovery nor Latent Labs has opened a wet lab, though perhaps they are generating large amounts of data at cloud labs like [Adaptyv](#). But the range of approaches is itself the interesting part. Given how rapidly and simultaneously these labs converged on similar technology (some data-heavy, some not) I am tempted to [side with Demis Hassabis that data was not the problem](#), at least to reach the current capabilities of these models.

There is another point that is very interesting in the paper by the Baker Lab. After running the whole design experiments and wet lab validation, they reran the predictions with the, back then, recently open-sourced AlphaFold 3, and observed that AF3 was an extremely powerful predictor of what designs were true binder versus which ones were not (AUROC 0.86). This is perhaps the most interesting point in this whole discussion. [When AlphaFold 2 was announced](#), I claimed that immune proteins appeared to be one of the hardest tasks because [coevolutionary information](#) was intrinsically not helpful due to the [recombination](#) and [somatic hypermutation](#) that these proteins experience. Indeed, for a long time all the structure prediction models have significantly struggled with antibody prediction and (even more) with antibody-antigen prediction. However, it seems that something in the new AlphaFold 3 model has endowed the model with the ability to better understand antibody-antigen interactions. I wish I could pinpoint the reason, but this would require a large number of ablations that I would never have the compute to achieve. My intuition is that it has something to do with the model's exposure to non-protein information, such as small molecules, forcing it to learn some of the physics that govern the interaction as well as the standard pattern matching.

If I had to bet, I would expect most of these companies to be exploiting AlphaFold 3-class models that are jointly modelling structure and sequence *i.e.* they can predict a structure from a sequence, but they can also generate a sequence from a structure, like ProteinMPNN, as well as generating both at the same time. These models will be based on transport-based generative approaches (like diffusion, or flow matching) and use similar loss functions to AlphaFold 3. I am sure there are a million small technical tricks, and certainly clever training recipes, but on the prior of having four independent groups arriving to this roughly at the same time, I would be happy to bet that this is a large portion of their success. I would also be willing to bet that a small but elite team with 5-10M USD (a pittance with today's AI seed rounds!) could likely get something on a similar level to what these guys have put out. After all, big portions of the architecture are public, and several open source models (see [BoltzGen](#) and [Germinal](#)) have achieved performance not far behind their well-funded competitors.

4. This will be important later, when we talk about the business models. But, for now, let's return to the original question: are these models going to help us design new drugs?

From antibody binder to antibody drug

Antibody *binders* are very different from *antibody drugs*.

The non-life-sciences world tends to assume the hard problem in drug discovery is designing the drug. It is not. We have been taking antibodies to the clinic with enormous success: there are some 200 FDA-approved antibodies, with over 1,500 at some stage of the clinic, and with the recent success of antibody-drug conjugates (ADCs) and other novel modalities of antibodies, there will only be more and more in the next few years. Finding antibodies has never been the problem. You can search online for a few dozen companies that run hybridoma or phage display analysis at very competitive prices, with very reasonable success rates. The problem is converting these binders into drugs.

First of all, to be a successful drug, **the binder has to become developable**, which is a polite word for a long list of ways a molecule can betray you. It has to be **stable enough** to survive months in a vial without unfolding or losing potency, and to tolerate the freeze-thaw cycles and the heat excursions of a real supply chain. **It has to stay a monomer**: antibodies that quietly aggregate are both less effective and more dangerous, because aggregates are one of the things the immune system likes to notice. It has to **express well in CHO cells** at a titre that makes manufacturing economic, fold correctly, and purify cleanly; **a beautiful binder that yields a hundred milligrams per litre instead of several grams is a binder that costs too much to make**. It **cannot be too hydrophobic**, or it sticks to columns and to itself; it cannot be polyreactive, latching nonspecifically onto unrelated proteins, or you inherit toxicity and a half-life problem. **Then there is immunogenicity**: a fully human-looking sequence that nonetheless carries a T-cell epitope can provoke anti-drug antibodies that neutralise the therapy outright.

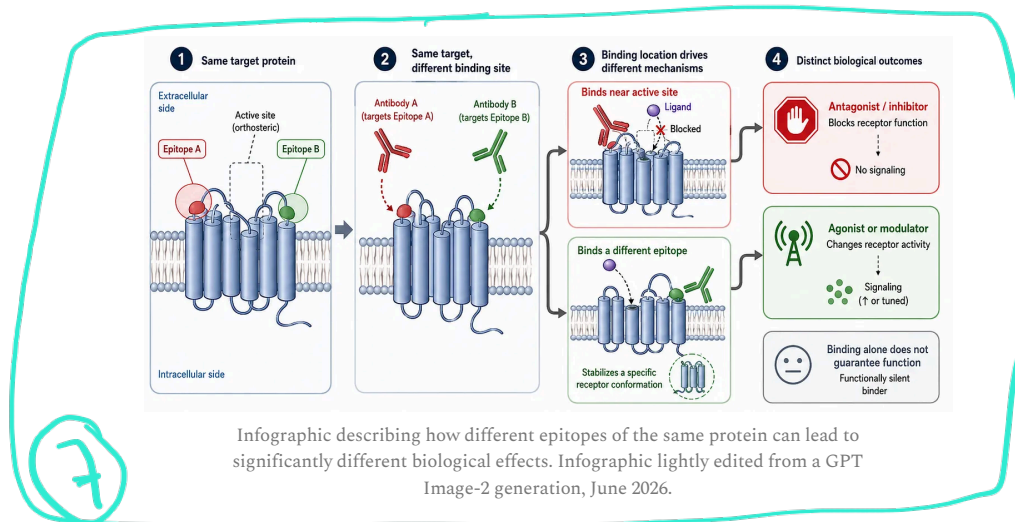
Beyond developability there are a million other problems related to how the antibody drugs perform inside a living body. For example, one is half-life and pharmacokinetics *i.e.* whether the molecule actually lingers in circulation long enough to dose monthly rather than daily, which is mostly a question of how it engages the [FcRn recycling pathway](#) and how fast the kidneys clear it. Tissue and cross-reactivity are also problems, such as when the antibody binds to some other things that you could not even have imagined. What is more, **the further you advance into the preclinical process, the properties that matter become slower and more expensive to validate**.

The argument that has been floated around is that, if antibodies are *better from the get go*, these companies will accelerate the preclinical phase, leading to more shots on goal to reach the clinic. There are some early reasons to be interested in this. If you read the [Latent Labs whitepaper](#), the majority of the antibodies they designed seem to have some excellent developability properties: the stability, monomericity and yield are comparable to therapeutic antibodies and frankly much better than most first-round designed antibodies; although the hydrophobicity and polyreactivity still need some work. Similar metrics are reported across other companies' papers.

I agree that these methods will sometimes speed up some parts of the preclinical process. However, **the real problem is whether the antibody produces a real benefit in a real patient, which is a property not of the molecule but of the biology**, and which no amount of binding affinity or easily assayable properties can buy you. Of the 88% of antibodies (roughly 7 in 8) that fail in the clinic, **every single one of them passed the filters that these companies claim to be able to overcome, and even demonstrated efficacy in advanced animal models, yet failed for a completely different reason**.

One of the best examples here are the anti-amyloid antibodies developed in the never-ending quest for cures of Alzheimer's disease. Take [solanezumab](#): it was an extremely good antibody by all quantitative metrics, binding monomeric β -amyloid at picomolar affinity, and sailing through any developability panel you could possibly imagine, yet **it failed its Phase III trials outright**. In contrast, [aducanumab](#) bound the same target some ten thousand times more weakly (micromolar, which these models, and frankly most of the people I know, would discard), and yet it was the first in the class to **show a human signal**, because it was able to bind the aggregate rather than the monomer. But,

and here is the catch, **both antibodies failed to produce a meaningful clinical effect** because the biological hypothesis (that reducing amyloid deposits would halt disease progression) **doesn't seem to hold**. And this isn't limited to diseases like Alzheimer's, where the translational difficulties are notorious; there are others, in antibodies (**afelimomab** in sepsis) and in drug discovery generally.



This does, however, point to a very attractive advantage of *de novo* design: **the ability to control the epitope of the antibody**. In short, since proteins are three-dimensional machines, **where an antibody binds matters as much as whether it binds**. Two antibodies with identical affinity for the same protein can have opposite biological effects: one lands next to the active site and shuts the protein down, whereas the other binds a harmless patch elsewhere and does nothing. For some targets this is extremely important. Take G protein-coupled receptors (GPCRs), the target of nearly a third of all approved drugs and a notoriously hard one for antibodies. Indeed, Nabla.Bio **showed their ability to design many antibodies against this target class**, including validated agonists and antagonists *i.e.* they didn't just bind, they elicited a biological effect.

In summary, **the hard truth in the drug discovery industry is that most drugs fail even in the best conditions**. Every single drug that went into clinical trials worked on the animal models, was approved by the key opinion leaders, and **ticked every box to convince the drug developer to invest tens of millions of dollars into clinical trials**. As they say, if cancer in humans was like cancer in mice (or even in primates), we would have cured most of it already. The ultimate hard truth about drug discovery is that the bottleneck is human studies. An enormous graveyard of extremely promising drugs sits behind us, none of which went anywhere.

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Is this a business?

We have concluded that there is a lot of good science in this two-year revolution, but that there is (or rather, I have) some skepticism about their ability to truly transform

drug discovery. The next potential question is: do these companies have a business?

The four companies alluded to in the introduction are the near definition of [platform companies](#): they are working on a technology engine that can generate *multiple* drugs. Because of the nature of the pharmaceutical industry, where all of the revenue comes from a few extremely profitable drug launches, these companies end up with two broad strategies: they either *make a drug* (what I will call the “biotech” model), or *help someone else make a drug* (what I will call the “outsourcing” model).

The outsourcing model is very attractive: you find a few Big Pharmas, and get them to sign you a *biobucks* deal to develop drugs for them. What this means is that the pharma company will pay an *upfront* amount for usage of the company; if you are a hot company, this upfront can range from the mid-double-digit millions to the low-triple-digit millions. But that is not all, the pharma will also give you *milestone* payments, which means that everytime that the molecule goes through some value inflection point (say, IND approval, or first injection in a Phase I trial), the Pharma will wire an even larger amount of money. You will also have access to *royalties*, which means a share of the revenue produced by the sales of the molecule if/when it reaches approval. These humongous amounts of money will often be reported, in a headline, as an even more humongous number that contains every possible payment. [Whether these humongous numbers will actually materialise](#) is an entirely different story, but even the cash upfront can be an outrageous amount of money for an otherwise cash-starved biotech company.

Here is the deal: if any single company had created this technology, they would be the sweetheart of the pharmaceutical industry. When four companies reach the same milestones within months, then their ability to capture value is reduced because the technology nears [commoditisation](#). A pharma company shopping for antibody frontier models now has four doors to knock on instead of one, and that competition compresses margins. I should caveat, however, that Chai Discovery has performed particularly well here, because their announced partnerships with [Eli Lilly](#) and [Pfizer](#) report *recurrent revenue* from access to this. While they don't get to report the very humongous number of the total deal value, the disclosed Eli Lilly fee is reportedly in the eight-figure range. It remains to see, however, whether they will be able to maintain these deals in the light of increased competition.

Commoditisation is a challenge to any industry, but it is even worse in the software world because open source is racing behind. Within months of the commercial releases, the open source [BoltzGen](#) pipeline was published with the ability to generate nanobodies (note: while editing this article, the [Boltz team](#) announced [BoltzProteo](#), which is an upgraded version of this pipeline with a better model; their whole thesis is to be the default model provider in the commoditisation era, and they are delivering like they mean it). Okay, you may argue that they are using [Boltz-2](#), which was trained on Recursion's [BioHive supercomputer](#), so they are not really your garden variety academic group. But you can also get [Germinal](#), a model trained by academics at Stanford and the Arc Institute. While these guys have access to quite a lot of compute for an academic group, their method is really simple: they backpropagate through AlphaFold Multimer with a mix of gradients from AF-M itself, and an antibody language model (IgLM); then they map the structure to a sequence using AbMPNN and verify with AlphaFold 3. This, plus a few tweaks from the team (and these are all really simple things, like constraining the

generation to avoid certain secondary structures) is enough to deliver double-digit success rates in the nanomolar range.

If the outsourcing, “B2B SaaS” play is not an attractive one, the alternative strategy is for these companies to play the biotech model. Here, they would **pick some valuable targets, develop some antibodies against them, push them through the drug discovery pipeline, and enter the clinic as soon as possible.** This very much seems like what Xaira (and, I would bet, at least some teams within Isomorphic Labs) are doing in the background; of course, we do not hear that much about them because, [having raised billions of dollars](#), they don’t really have to worry about making announcements for investors and can focus on the long-term delivery.

The problem is that asset development is *fundamentally* a terrible business model. After all, a drug that hits the clinic has a 88% probability of *failure* (that’s because it’s an antibody, a small molecule would have a whopping 92%). That is, I will not get tired of saying it, a byproduct of years of work that has shown success in multiple animal models and any possible experimental test before taking it into humans. Despite the absurdly large amounts of money that some drugs generate, [the pipelines of most pharmaceutical companies return less than the cost of capital](#). And yet they have dozens of drugs entering the clinic every year, and can diversify the risk, so what hope has a small biotech? When the attrition-adjusted cost of taking a drug to approval is about \$2 billion dollars, even extremely well-funded companies like Xaira (\$1B seed) and Chai (\$225M raised, rumoured to be raising more) may be short-changed.

There is only one way to win, really. They need to **pick one differentiated clinical candidate, prove it, and then let it fund the platform.** But for the reasons that I have mentioned before, they can’t afford to go against a random target. They need to **pick some target that is sufficiently biologically well-validated that they minimise the risk of clinical trials going awry**, or otherwise they face the same inefficiencies as most pharmaceutical companies. At the same time, they cannot afford competition, so they need to focus on something that no other competitor can do: something that their platform unlocks. The only one of the companies that seems to be doing this is Nabla, with their published work [developing antibodies for G-protein coupled receptors](#) or the previously discussed [peptide-MHC antibody mimics for KRAS](#).

There can be a lot of discussion of what constitutes a great target for one of these companies, but notice that the two requirements pull in opposite directions. The biology has to be well-trodden enough that the target is not the gamble, but the particular way the molecule is looking at it. Likewise, the molecule has to be one that only *de novo* design could have produced, which by definition pushes you toward the targets classical discovery never cracked. The ideal candidate is therefore a strange chimera: a target safe enough that a small biotech can survive its clinical trial, and hard enough that nobody could have made the antibody any other way. The success is clear: a success like Revolution Medicines’ [daraxonrasib Phase III results](#), **finding a novel way to succeed with KRAS, a drug target known for fifty years that however remained uncracked.** Unfortunately, the intersection is narrow, and everyone capable of fishing in it is fishing in the same pond.

So here is the open question for these antibody drug discovery companies: can they deliver a differentiated clinical candidate? I will be very interested to see what they come up with.

The data play

There is a counterargument I owe an honest hearing, because it is the best one the believers have: the value of data in delivering an exponential improvement in capabilities.

[Platform biotechs](#) are not new; in fact, some of the great pharmaceutical successes of the last fifty years were built this way ([Amgen](#) on recombinant proteins, [Gilead](#) on antivirals, [Regeneron](#) on humanised antibodies, etc). What is supposed to make this wave different is the **data flywheel**. In a traditional biotech, a failed project is killed and you learn almost nothing, but in an AI biotech, a failed project becomes valuable training data. Every campaign, win or lose, makes the model better. The moat compounds, and the frontier pulls away from the open-source competition trailing it by only a few months. If you believe that, then the commoditisation worry I just raised should just be temporary, and the company that generates the most data wins.

The main argument against the data flywheel in the antibody field is that the data that is most valuable is also the most expensive to get. At every step of the preclinical journey, the problem increases in value and cost: finding binders is relatively easy, optimising them for developability is a bit harder, getting the pharmacokinetics is harder still, animal experiments are more expensive the closer you get to human, and testing biological hypotheses in a human trial is a multi-million dollar experiment that so far has no real alternative. If the data you need to power the flywheel is at the top of the value chain, as seems the case, then you are in trouble.

The question to always keep in mind is: **does the marginal data get cheaper than the marginal value it unlocks?** If the data required to keep improving the models grows faster than the improvement it buys, then the flywheel does not run away from the competition. The equilibrium point is somewhere around where the cost of the next accuracy increment exceeds the industry's willingness to pay.

In this model, the only way to make the flywheel work is to build an advantage at the very early parts of the discovery. Here is where I think Nabla.Bio's strategy is really clever: their focus is on modalities that have so far been inaccessible but the biological interest is clear, such as the GPCR modulators and the pMHC mimetics. Their [JAM platform](#) combines *de novo* design with what amounts to neighbourhood expansion: generating large amounts of data around a promising binder with a robotised pipeline, essentially. Because their data is closest to physics binding, it is unsurprising that their platform gets intricate control on the binding mode of the antibodies and enables some of their breakthroughs.

The LLM world offers a clear parallel. In 2026, it is no longer controversial to say that [pure scaling by training on data is largely considered stalled at most frontier labs](#). The scaling these days has taken three forms: reinforcement learning at scale, generally with [verifiable rewards](#); post-training on specific tasks (such as the many white-collar datasets collected by companies like Scale AI and Mercor); and test-time scaling, which is just letting the models think for longer. These do not have a clear parallel in biology. While there are some reports of test-time scaling in biology, most of them can be summarised in running an inference process for longer (e.g. more diffusion samples or more recycling steps). Reinforcement learning is far harder in biology, because we can't cheaply (or at all) simulate outcomes, so the reward signal has to come from the

wet lab, where it's slow and expensive. Therefore, if you agree as we have posed that more pretraining will give you diminishing returns (as the [scaling law](#) foretells), the only play left is post-training: focusing on specific problems and moving forward. But post-training is a lot about becoming very good on specific assets, and less about getting an overly powerful model that takes forward the data flywheel.

I believe in the value of the data flywheel in AI for life sciences with all my heart. So much so that I have dedicated my career to it. However, unless the strategy focuses at the preclinical end, where the data is cheap and the models can accrue value, I do not think it makes sense.

Conclusions

The developments in *de novo* antibody discovery that we saw in 2025 and 2026 are one of the most compelling advances I have seen in the field. They remind me a lot of AlphaFold 2 in 2021; not on the magnitude, but on the amount of proof delivered. Having spent a lot of time over the last months thinking about it, I have concluded that there is no catch... in the science.

The catch is in the business model. There are many questions about how well these model companies will perform as a business. In general, I am skeptical that the platform-as-a-service model will work here. I think we are likely to see a scenario mirroring LLM commoditisation, where the open source model tracks closely behind the frontier. This time, unlike natural language models, there will be no clear reinforcement learning loops or additional training techniques; and, if the data required to increase the accuracy of the models needs to grow exponentially while the cost of data acquisition stays constant, models will halt as soon as the cost of developing them surpasses what the industry is willing to pay for them.

The open question for the field is whether these companies will be able to play the platform biotech game. In almost all cases, this will require them to commit to at least one drug candidate where their technology truly differentiates them. If this candidate(s) fare well, the revenue they generate may power the rest of the technology. Otherwise, we may see another wave of AI for drug discovery companies struggle.

The revolution happened; will the business sustain it?

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